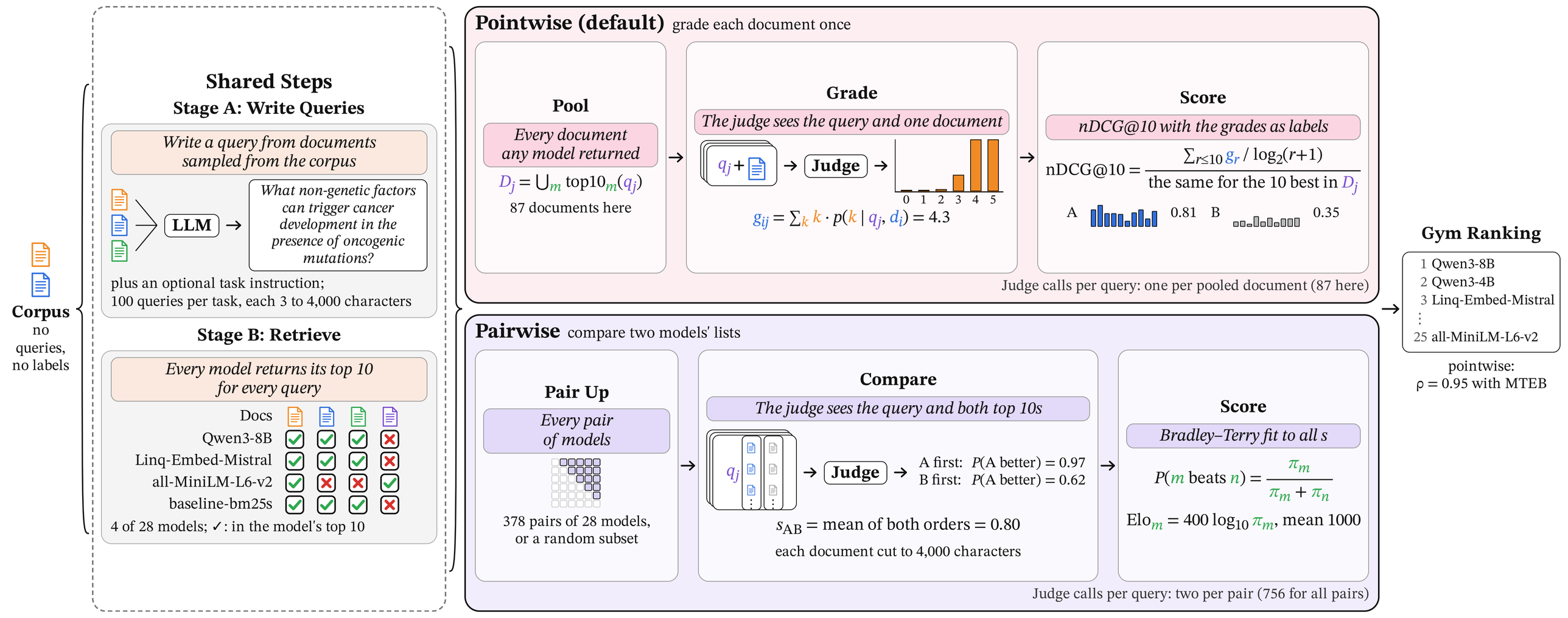

Corpus
no queries, no labels
Shared Steps
Stage A: Write Queries
Write a query from documents sampled from the corpus
LLM
What non-genetic factors can trigger cancer development in the presence of oncogenic mutations?
plus an optional task instruction;
100 queries per task, each 3 to 4,000 characters
Stage B: Retrieve
Every model returns its top 10 for every query
Docs
Qwen3-8B
Linq-Embed-Mistral
all-MiniLM-L6-v2
baseline-bm25s
4 of 28 models; ✓: in the model's top 10
Pointwise (default) grade each document once
Pool
Every document any model returned
$D_j = \bigcup_m \text{top10}_m(q_j)$
87 documents here
Grade
The judge sees the query and one document
q_j + → Judge →
0 1 2 3 4 5
$g_{ij} = \sum_k k \cdot p(k \mid q_j, d_i) = 4.3$
Score
nDCG@10 with the grades as labels
$\text{nDCG@10} = \frac{\sum_{r \le 10} g_r / \log_2(r+1)}{\text{the same for the 10 best in } D_j}$
A 0.81 B 0.35
Judge calls per query: one per pooled document (87 here)
Pairwise compare two models' lists
Pair Up
Every pair of models
378 pairs of 28 models, or a random subset
Compare
The judge sees the query and both top 10s
q_j → Judge →
A first: P(A better) = 0.97
B first: P(A better) = 0.62
s_{AB} = mean of both orders = 0.80
each document cut to 4,000 characters
Score
Bradley–Terry fit to all s
$P(m \text{ beats } n) = \frac{\pi_m}{\pi_m + \pi_n}$
$\text{Elo}_m = 400 \log_{10} \pi_m$, mean 1000
Judge calls per query: two per pair (756 for all pairs)
Gym Ranking
1 Qwen3-8B
2 Qwen3-4B
3 Linq-Embed-Mistral
25 all-MiniLM-L6-v2
pointwise:
ρ = 0.95 with MTEB